

NIKHIL CHEEMALA

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PROFESSIONAL SUMMARY

Backend and Generative AI engineer building LLM applications, RAG systems, agentic workflows, and production-oriented APIs, with a strong foundation in software engineering, retrieval systems, and backend architecture.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL

AI / Generative AI: LLMs, RAG, LangChain, LangGraph, LLM Tool Calling, Agentic AI, Prompt Engineering, OpenAI, Gemini, Hugging Face

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Pydantic, Zod, JWT, Microservices

Frontend: React, Next.js, Tailwind CSS, Redux, Redux Toolkit, React Query

Databases: MongoDB, MongoDB Atlas, PostgreSQL, pgvector, ChromaDB, Redis, Mongoose

Cloud & DevOps: Docker, CI/CD, Vercel, Render, Git, GitHub, Postman, Playwright, n8n, Zapier, Claude Code, OpenAI Codex, GitHub Copilot

PROJECTS

Enterprise RAG Platform

[GitHub](#)

Tech: RAG, Hybrid Retrieval, BM25+, RRF, Neural Reranking, Access Control

- Built a **production-oriented RAG platform** for heterogeneous enterprise documents with grounded answers and source citations.
- Engineered dense + BM25+ hybrid retrieval with **RRF fusion, neural reranking**, access control, and RAG security guardrails.
- Achieved 89% Recall@5, 89% Recall@10, and **0.83 MRR** in the initial retrieval evaluation.
- Measured 89% answer grounding with **930 ms retrieval latency**, establishing a baseline for optimization.

AI Travel Booking Agent

[Live](#) | [GitHub](#)

Tech: FastAPI, LLM Tool Calling, Agentic AI, Pydantic, OpenAI/Gemini

- Built a **conversational agentic travel planner** that converts natural-language requests into structured trip workflows using LLM planning and session state.
- Engineered registry-based tool calling, validation, and bounded retries for resilient **multi-turn correction handling**.
- Improved slot extraction from 77.6% to 91.6% across 143 field-level comparisons; complete-record extraction rose to **91.4%** across 35 scenarios.
- Improved workflow success from **88.9% to 97.2%** across 36 HTTP scenarios and correction handling from 70% to 100% across 20 scenarios; p95 latency was 32.3 ms with 0/45 request failures.

Developer Jobs & Knowledge Publishing Platform

[GitHub](#)

Tech: Next.js, TypeScript, Tailwind CSS, MongoDB Atlas, Mongoose

- Built a **full-stack developer platform** combining job listings, technical publishing, interview preparation, and system-design content.
- Implemented an admin CMS for content lifecycle management, Markdown editing/preview, and **search, categories, tags, and SEO fields**.
- Added protected admin routes, signed HTTP-only sessions, and **rate limiting with Zod validation**, with Vercel deployment.

EDUCATION

VNR VJIET, Hyderabad | B.Tech in Computer Science and Engineering | 2022–2026 | **CGPA: 8.33**

ACHIEVEMENTS

- LeetCode: Peak rating 1829; 400+ problems solved.
- CodeChef: 3-Star Programmer; peak rating 1631.
- Smart Interviews: Diamond Certified Smart Coder; Global Rank 300.